

EEE4114F Final Exam

June 2024

Duration: 3 hours

Total: 80 marks

Empl ID: _____

Instructions:

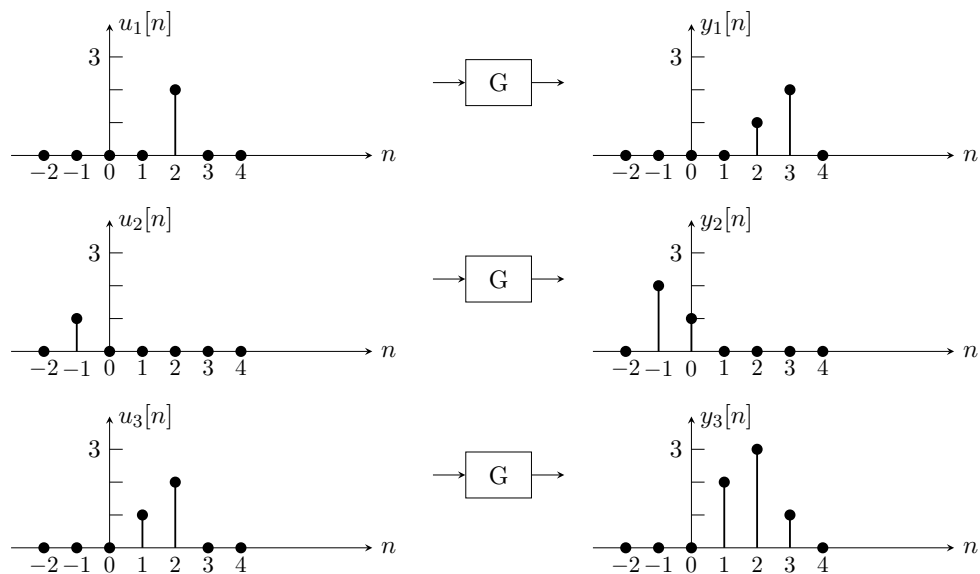
- This is a closed-book exam.
- There are two parts to this exam.
- **Part A** has 5 questions totalling 40 marks. You must answer all of them.
- **Part B** has 3 questions totalling 40 marks. You must answer all of them.
- Both parts can be answered in the space provided on the exam script, and you can write on the back of the printed pages if required. Answer books will be available should you need additional writing space.
- You must use a pen to complete the exam — pencil will only be marked for drawings. Values indicated on drawings must be written in pen. Please show all of your working clearly — method and approach matter.
- A table of standard Fourier transform and z-transform pairs appears at the end of this paper.
- You have 3 hours.

Question	Marks	Grade
1	5	
2	10	
3	10	
4	5	
5	10	
6	15	
7	20	
8	5	
Total:	80	

Part A: Digital Signal Processing

Question 1 [5 marks]

The system represented by G in the figure below is assumed to be time-invariant:



When the inputs to the system are $u_1[n]$, $u_2[n]$, and $u_3[n]$ the outputs are $y_1[n]$, $y_2[n]$, and $y_3[n]$ as shown. Show that the system is not linear.

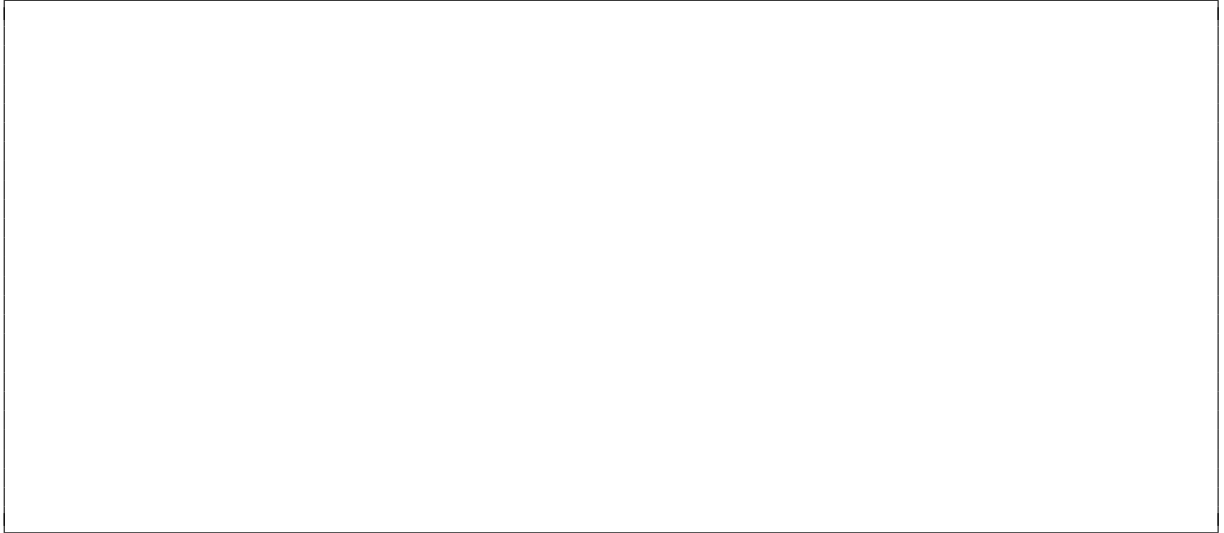
Question 2 [10 marks]

A LTI system has a step response

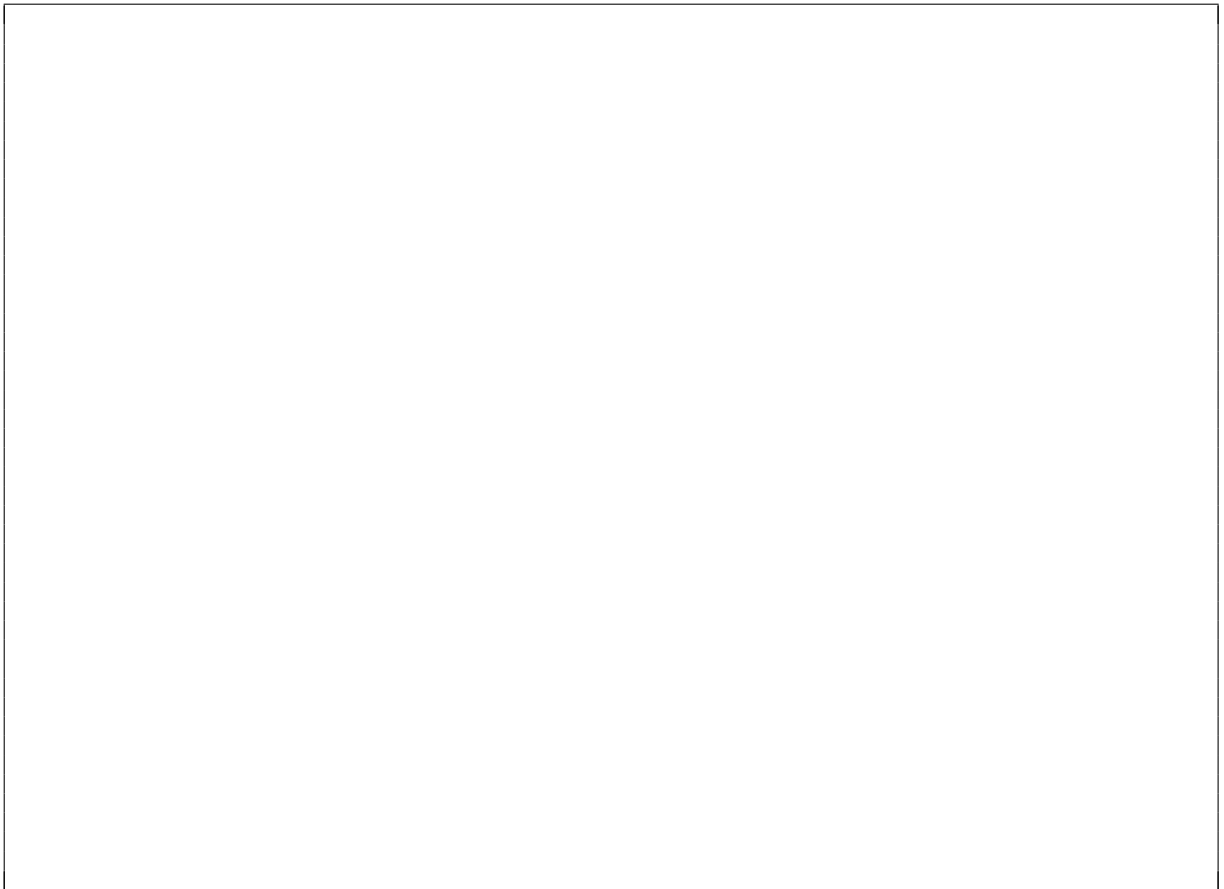
$$g[n] = n \left(\frac{1}{2} \right)^n u[n].$$

In other words, when the input is $x[n] = u[n]$ then the output is $y[n] = g[n]$ above.

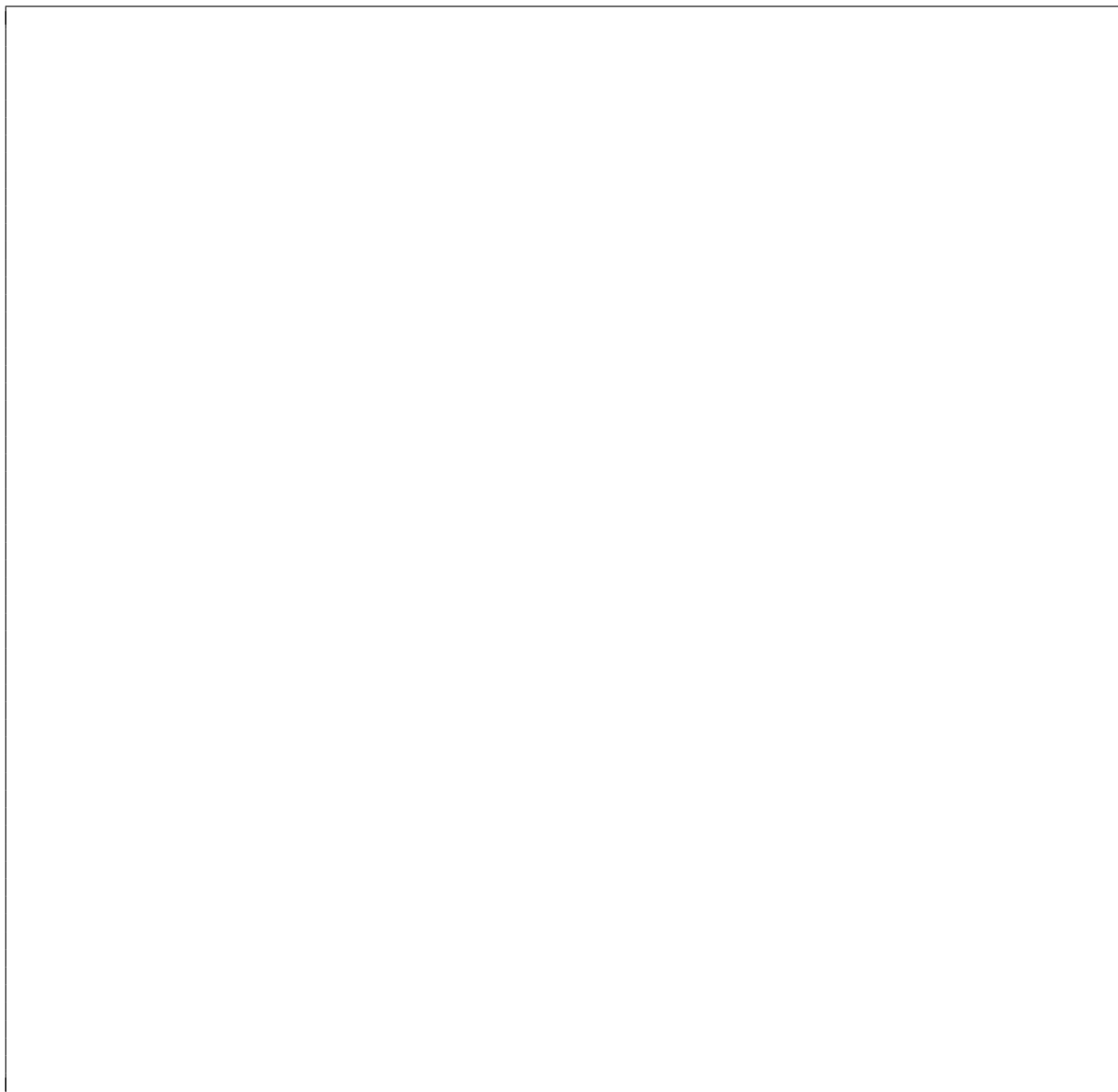
1. Plot $g[n]$ over the range -4 to 4 .



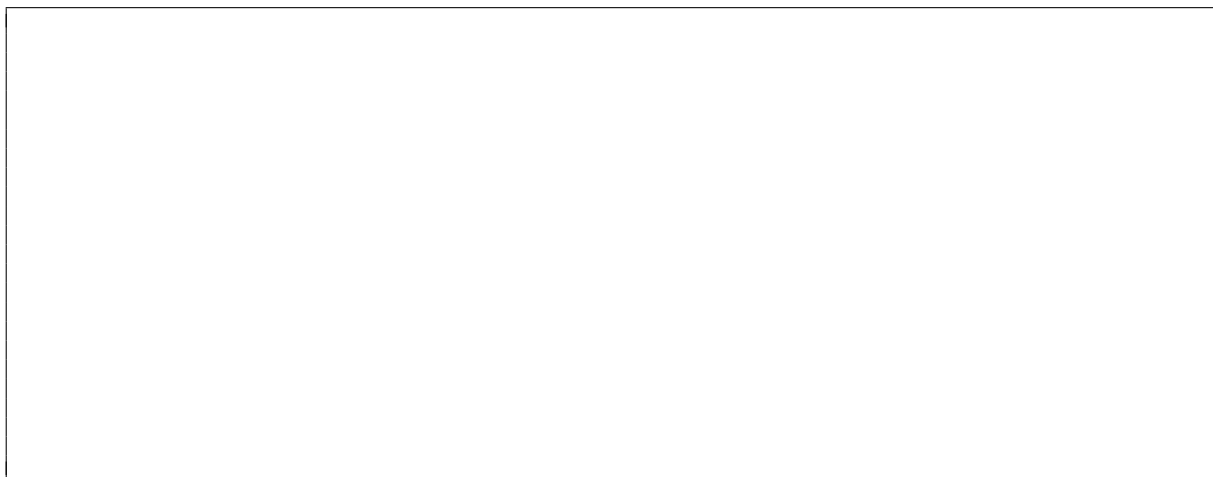
2. Use the DTFT pair $a^n u[n] \xleftrightarrow{\mathcal{F}} (1 - ae^{-j\omega})^{-1}$ along with the frequency differentiation property to find the Fourier transform of $g[n]$.



3. Find and plot the system impulse response over the range -4 to 4 . You may want to use the fact that $u[n] - u[n - 1] = \delta[n]$.



4. Is the system causal? Why?



Question 3 [10 marks]

A discrete-time, causal LTI system produces the output

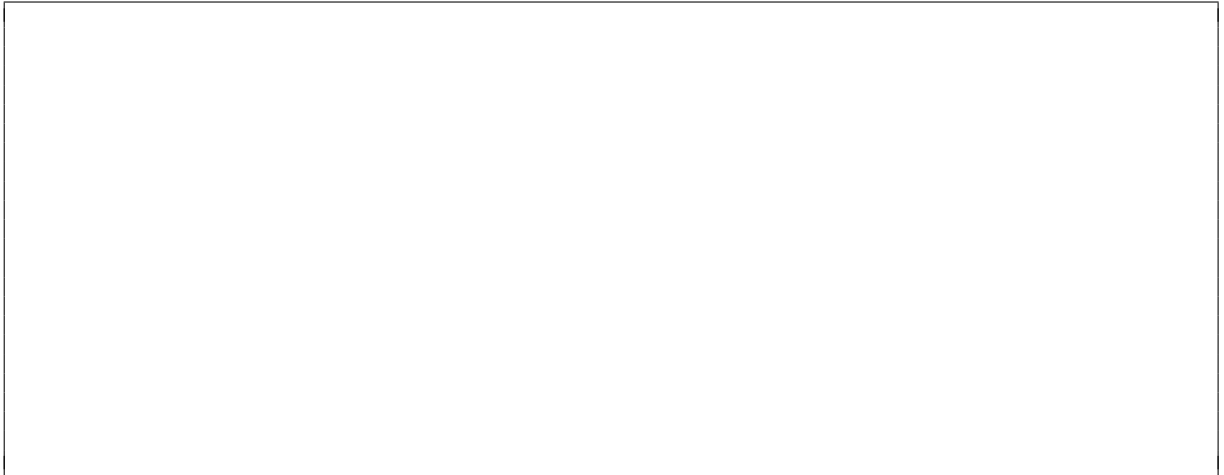
$$y[n] = \delta[n] + 2\delta[n-1] + \delta[n-2]$$

in response to the input

$$x[n] = \delta[n] + \delta[n-2].$$

1. Find the system function and sketch the pole-zero plot and the ROC.

2. Is the system stable? Why?

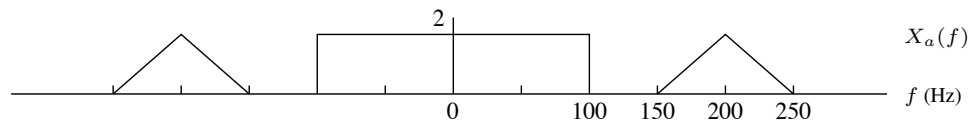


3. Find the impulse response of the system and sketch it over the range -4 to 4



Question 4 [5 marks]

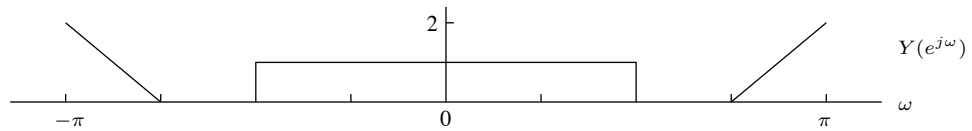
1. A continuous-time signal $x_a(t)$ has the Fourier transform shown below:



Let $x[n] = x_a(nT)$ be the sampled version of $x_a(t)$ with sampling period T , and let $X(e^{j\omega})$ be the DTFT of $x[n]$. Sketch $X(e^{j\omega})$ for $-\pi \leq \omega \leq \pi$ when

1. $T = 1/500$ seconds
2. $T = 1/400$ seconds.

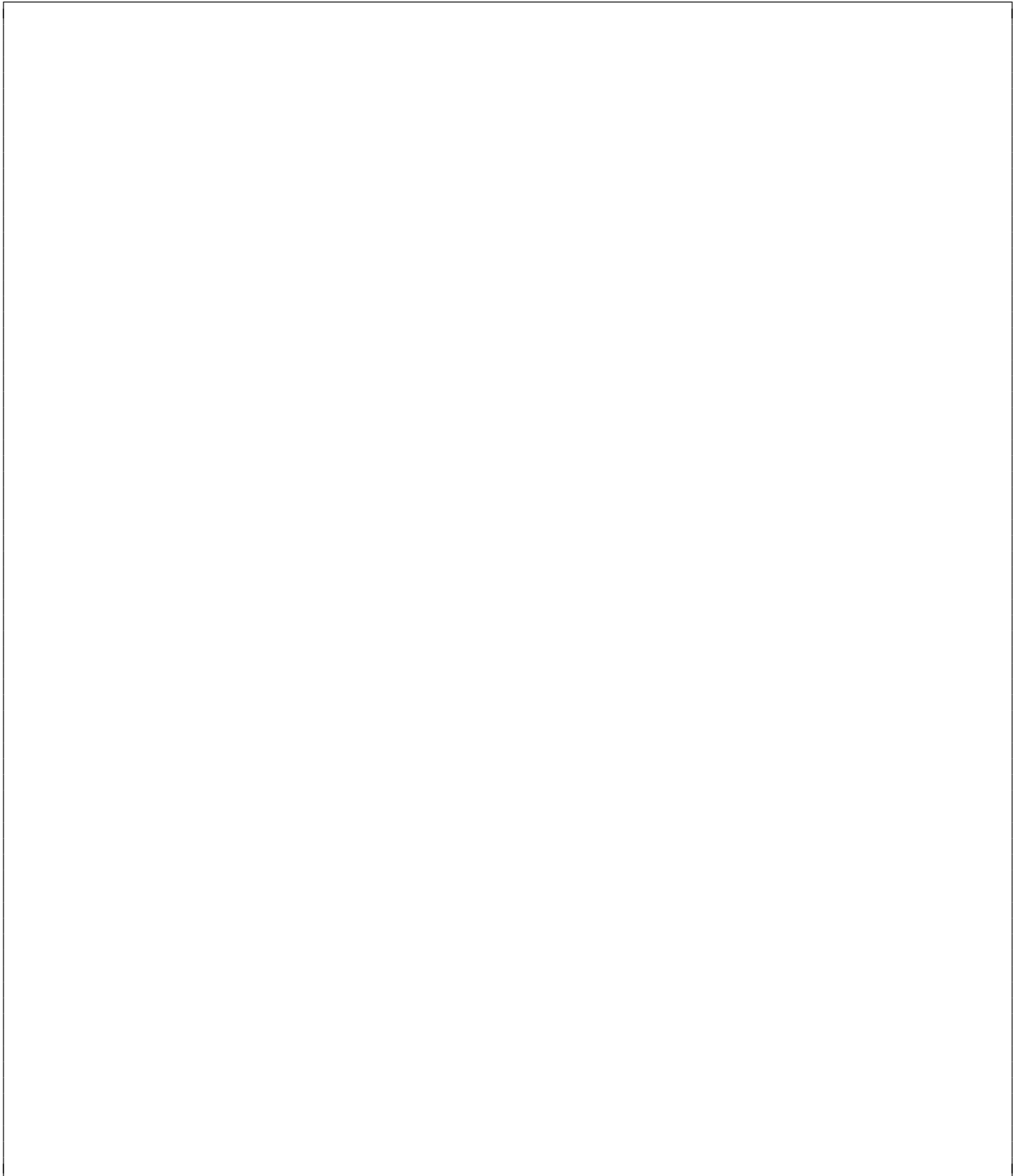
2. A signal $y[n]$ has the DTFT below:



The expanded signal $y_e[n]$ is defined as

$$y_e[n] = \begin{cases} y[n/2] & n \text{ even} \\ 0 & n \text{ odd} \end{cases}$$

Plot the DTFT $Y_e(e^{j\omega})$ of $y_e[n]$.



Question 5 [10 marks]

Answer the following:

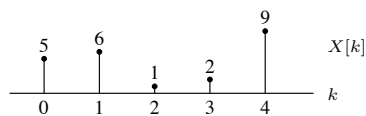
1. The even samples of the DFT of a 9-point real signal $x[n]$ are given by

$$X[0] = 3.1, \quad X[2] = 2.5 + 4.6j, \quad X[4] = -1.7 + 5.2j,$$

$$X[6] = 9.3 + 6.3j, \quad \text{and} \quad X[8] = 5.5 - 8.0j.$$

Determine the missing odd samples of the DFT.

2. The DFT of a 5-point signal $x[n]$ is shown below:

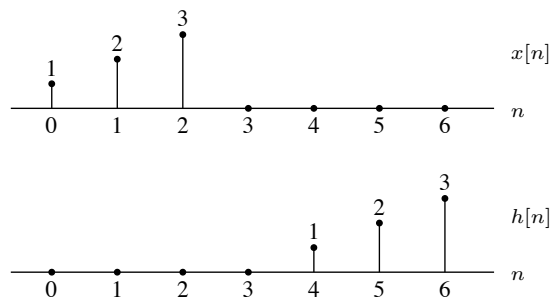


A new signal $g[n]$ is defined by

$$g[n] = W_5^{-2n} x[n], \quad 0 \leq n \leq 4.$$

Find and plot the DFT coefficients $G[k]$ of the signal $g[n]$, for $0 \leq k \leq 4$.

3. Find and plot the 7-point circular convolution of the following two signals:



END OF PART A

Part B: Machine Learning

Question 6: Fundamental Concepts [15 marks]

Suppose you need to develop a machine learning algorithm to estimate the number of people in a venue. This could be useful for applications such as efficient control of HVAC (Heating, ventilation and air-conditioning) systems to adjust temperature and ventilation strength based on the number of occupants. It could also be used by lecturers to determine estimates of student attendance in lectures without requiring strict attendance registers.

You decide not to use computer vision to record and detect people, as this poses ethical privacy concerns. Instead, you decide to set up a simpler sensor system that can be placed in a venue to collect data. The collected data is considered '*small*' and has been summarized in Table 1

Variable Name	Type	Min Value	Max Value	Units
n_occupants	Integer	0	200	count
CO2	Continuous	400	1400	ppm
sound_level	Continuous	40	90	dB SPL
temperature	Continuous	10	38	$^{\circ}C$
venue_size	Continuous	20	200	m^2

Table 1: A summary of the collected data

1. List all of the variables that would be considered input features. (2)

2. List all of the variables that would be considered targets. (2)

3. What is the purpose of feature scaling and would it be necessary for this dataset? (2)

4. Suppose you decide to train a linear regression model to estimate the target function. Write down the hypothesis function, $h_{\theta}(\mathbf{x})$, for this model if all input features are to be used and θ_0 is a bias parameter. (2)

Suppose the dataset was randomly sampled into one training set and one test set. Answer the following questions, and indicate the selected choice by marking the corresponding checkbox.

5. The parameters should be updated using _____. (1)
- ☐ Test data only.
 - ☐ Only training data.
 - ☐ Both training and test data.
 - ☐ It depends on the variance of the model.
6. Would the performance of the model on the test set be a reliable estimate of the model's generalization performance? (1)
- ☐ Yes.
 - ☐ No.
 - ☐ It depends on the variance of the model.
 - ☐ It depends on the complexity of the model.

Suppose cross-validation is performed instead of a single split of the data.

7. Given the cross-validation results in the form of a box-whisker plot as shown in Figure 1, would training on a combined training and validation set be a reliable method to improve the performance of the model (increase in r^2 score)? Explain your answer. (2)

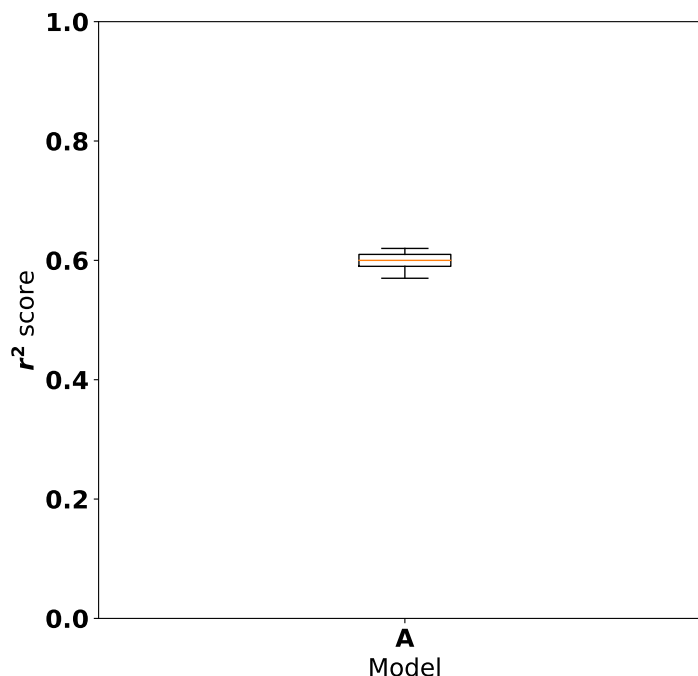


Figure 1: A box-whisker plot of the r^2 scores after cross-validation of the model.

[illegible]

8. Would regularization have a significant impact on model performance given the cross-validation results? Explain your answer. (2)

[illegible]

9. If you wanted to improve this model's performance what would you change? (1)

- ☐ Increase the number of input features
- ☐ Decrease the number of input features
- ☐ Change the order that the features appear in the input vector
- ☐ Perform additional splits of the data

Question 7: Neural Networks [20 marks]

Suppose you have a dataset, $\mathcal{D}$, for the problem in Q6. You decide to train an artificial neural network to solve the problem and you also decide to limit the input features to just CO2 and sound_level. The hidden layer uses relu activation functions and the output uses a linear activation.

The initial neural network parameters are:

$$\mathbf{W}^1 = \begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix}, \mathbf{W}^2 = \begin{bmatrix} 0.2 \\ 0.1 \end{bmatrix}, \mathbf{b}^1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, b^2 = 0$$

The gradients after the first forward pass have been calculated as:

$$\frac{\partial J}{\partial \mathbf{W}^1} = \begin{bmatrix} 0.023 & 0.011 \\ 0.027 & 0.014 \end{bmatrix}, \frac{\partial J}{\partial \mathbf{W}^2} = \begin{bmatrix} 0.18 \\ 0.19 \end{bmatrix}, \frac{\partial J}{\partial \mathbf{b}^1} = \begin{bmatrix} 0.05 \\ 0.1 \end{bmatrix}, \frac{\partial J}{\partial b^2} = 0.23$$

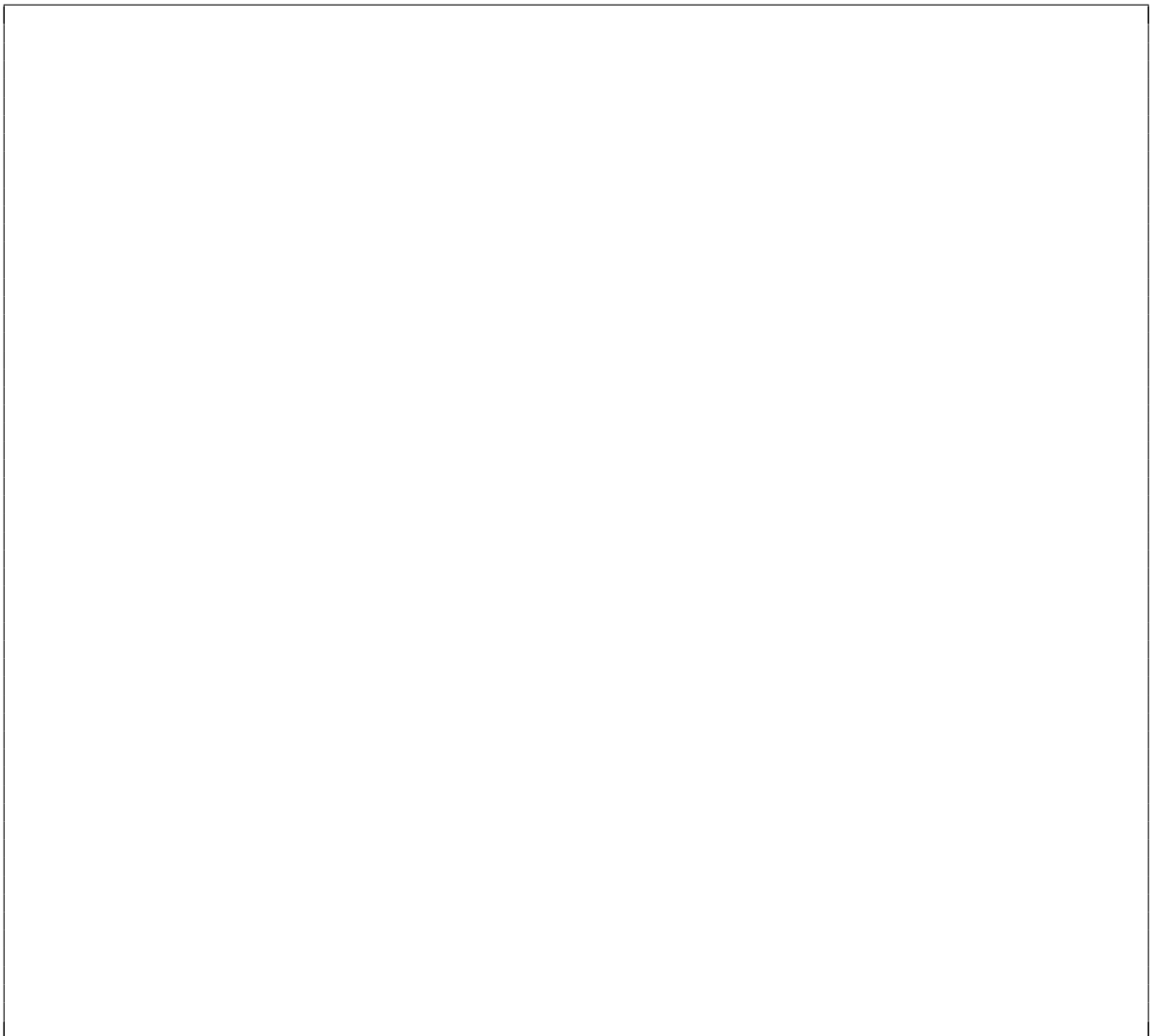
Answer the following questions:

1. How many neurons are there in the hidden layer of this neural network? (1)

2. Calculate the output of the neural network using the initial model parameters and the first input $\mathbf{x} = \begin{bmatrix} 0.5 \\ 0.6 \end{bmatrix}$ (4)



3. If the model parameters are updated after the first input example, what would the new model parameters be (2)
if the learning rate $\eta = 0.5$? Use the gradients provided in the question.



4. Calculate the output of the neural network for a second input $x = \begin{bmatrix} 0.3 \\ 0.4 \end{bmatrix}$. (3)

5. Calculate the mean squared error loss for the second training example if the target is $y = 0.2$. (1)

6. Given that: (4)

$$\frac{\partial J}{\partial w_{11}^1} = \frac{\partial J}{\partial a_1^2} \frac{\partial a_1^2}{\partial z_1^2} \frac{\partial z_1^2}{\partial a_1^1} \frac{\partial a_1^1}{\partial z_1^1} \frac{\partial z_1^1}{\partial w_{11}^1}$$

Determine the value of $\frac{\partial J}{\partial w_{11}^1}$ using the values calculated in part 4 as required.

7. If all of the input features were used from the dataset, would it make sense to use a convolutional neural network for this task? (1)

8. What makes convolutional neural networks more efficient for tasks such as image classification, compared to fully connected networks? Think about the number of input features and how this relates to the number of model parameters. (2)

9. If a given layer in a 2D CNN has a input with dimensions of $(64, 64, 1)$ and produces 10 feature maps, with kernels that have dimensions of $(5, 5)$, how many model parameters are there for that layer assuming bias terms are included? (2)

Question 8: Reinforcement Learning [5 marks]

Suppose an agent is placed in an environment as shown in Table 2a with a Q-table in Table 2b that has been developed through training an agent in a virtual environment. The agent can take actions from the following action-space $\mathcal{A} = \{\text{North, East, South, West}\}$ which causes it to move in the corresponding direction. These can be abbreviated N,E,S,W and the position of the Q value within a cell corresponds to the direction of the action. N is $\uparrow$, E is $\rightarrow$, S is $\downarrow$, W is $\leftarrow$. s_7 and s_2 are the terminal states and shaded in grey.

s_6 -1	s_7 2	s_8 -1
s_3	s_4	s_5 +3
s_0	s_1	s_2 -2

0.0	0.0	0.0
0.0 0.8	0.0 0.0	0.8 0.0
0.3 -0.1	0.0 0.7	0.5 -0.1
0.0 0.8	0.1 0.6	0.5 0.0
0.1	0.1	-0.2
0.6	0.8	0.0
0.0 0.4	0.2 -0.4	0.0 0.0
0.0	0.0	0.0

- (a) The agent's environment. Rewards are shown if they are non-zero. (b) The Q-table after an arbitrary number of episodes.

The Q-values are updated according to the following learning rule:

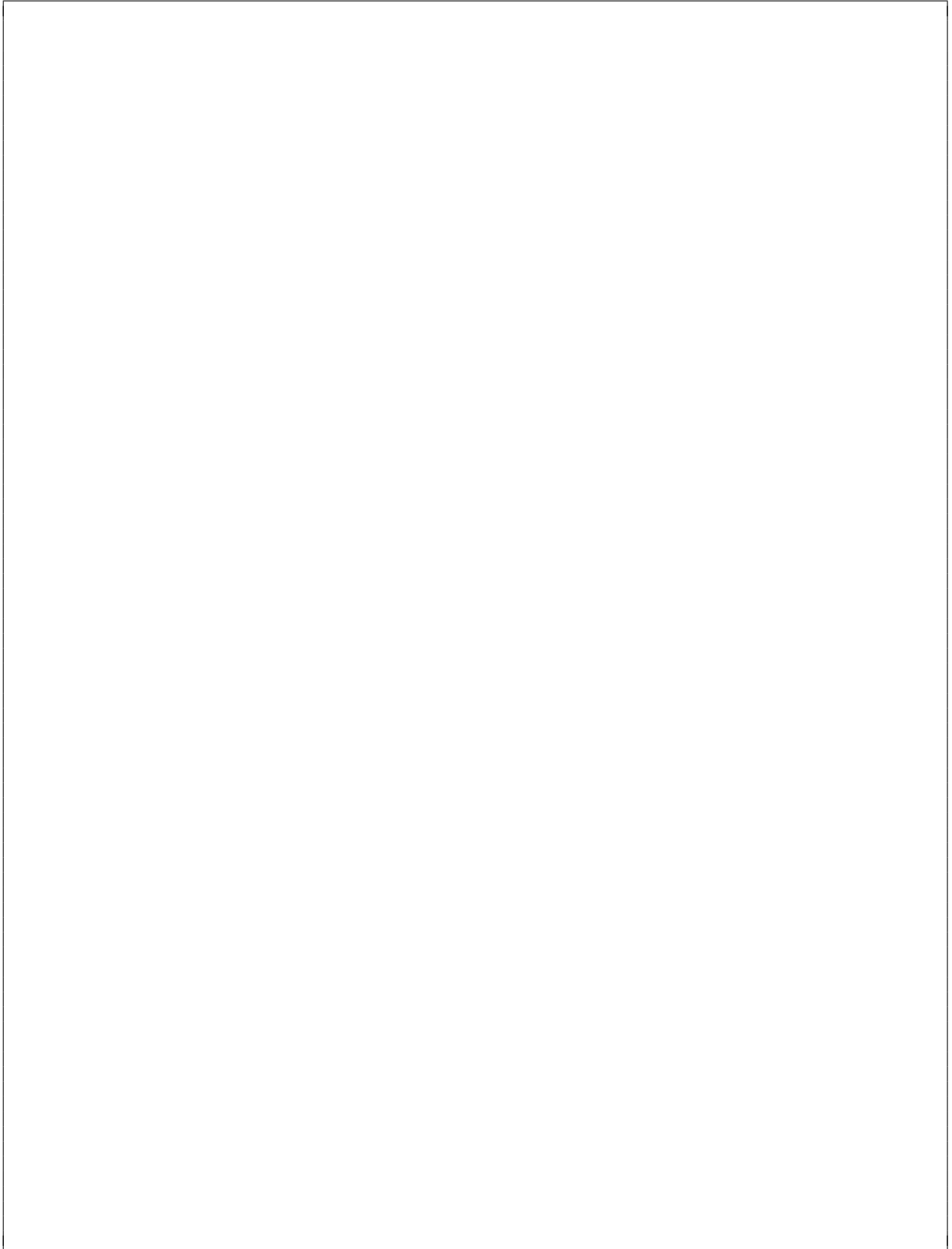
$$Q(s, a) \leftarrow Q(s, a) + \alpha [R(s') + \gamma \max_{a'} Q(s', a') - Q(s, a)]$$

Where $\alpha = 0.5$ and $\gamma = 0.8$. Answer the following questions.

1. Starting in the state, s_0 , write down the sequence of states, actions, and rewards in the form $(s_t, a_t, r_{t+1}) \rightarrow (s_{t+1}, a_{t+1}, r_{t+2})$ the agent will experience if it is using a greedy action policy, $\pi(s) = \arg \max_a q_\pi(s, a)$, to select actions based on the Q-table in Table 2b. (2)

2. If the policy used to train the model is an ϵ - greedy policy would it be an on or off policy? (1)

3. If the agent starts in state, s_0 , and chooses to go East twice, which Q-values would be updated, and what would their updated values be? (2)



END OF PART B

DSP information sheet

Transforms

$$X(j\Omega) = \int_{-\infty}^{\infty} x(t)e^{-j\Omega t} dt \quad \Longleftrightarrow \quad x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\Omega)e^{j\Omega t} d\Omega \quad (\text{CTFT})$$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \quad \Longleftrightarrow \quad x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega})e^{j\omega n} d\omega \quad (\text{DTFT})$$

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n} \quad \Longleftrightarrow \quad x[n] = \frac{1}{2\pi j} \oint_C X(z)z^{n-1} dz \quad (\text{ZT})$$

$$X[k] = \sum_{n=0}^{N-1} x[n]W_N^{kn} \quad \Longleftrightarrow \quad x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k]W_N^{-kn} \quad \text{with } W_N = e^{-j\frac{2\pi}{N}} \quad (\text{DFT})$$

Discrete-time Fourier transform properties

Sequences $x[n], y[n]$	Transforms $X(e^{j\omega}), Y(e^{j\omega})$	Property
$ax[n] + by[n]$	$aX(e^{j\omega}) + bY(e^{j\omega})$	Linearity
$x[n - n_d]$	$e^{-j\omega n_d} X(e^{j\omega})$	Time shift
$e^{j\omega_0 n} x[n]$	$X(e^{j(\omega - \omega_0)})$	Frequency shift
$x[-n]$	$X(e^{-j\omega})$	Time reversal
$nx[n]$	$j \frac{dX(e^{j\omega})}{d\omega}$	Frequency diff.
$x[n] * y[n]$	$X(e^{j\omega})Y(e^{j\omega})$	Convolution
$x[n]y[n]$	$\frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\theta})Y(e^{j(\omega - \theta)}) d\theta$	Modulation

Common discrete-time Fourier transform pairs

Sequence	Fourier transform
$\delta[n]$	1
$\delta[n - n_0]$	$e^{-j\omega n_0}$
1 $(-\infty < n < \infty)$	$\sum_{k=-\infty}^{\infty} 2\pi \delta(\omega + 2\pi k)$
$a^n u[n] \quad (a < 1)$	$\frac{1}{1 - ae^{-j\omega}}$
$u[n]$	$\frac{1}{1 - e^{-j\omega}} + \sum_{k=-\infty}^{\infty} \pi \delta(\omega + 2\pi k)$
$(n+1)a^n u[n] \quad (a < 1)$	$\frac{1}{(1 - ae^{-j\omega})^2}$
$\frac{\sin(\omega_c n)}{\pi n}$	$X(e^{j\omega}) = \begin{cases} 1 & \omega < \omega_c \\ 0 & \omega_c < \omega \leq \pi \end{cases}$
$x[n] = \begin{cases} 1 & 0 \leq n \leq M \\ 0 & \text{otherwise} \end{cases}$	$\frac{\sin[\omega(M+1)/2]}{\sin(\omega/2)} e^{-j\omega M/2}$
$e^{j\omega_0 n}$	$\sum_{k=-\infty}^{\infty} 2\pi \delta(\omega - \omega_0 + 2\pi k)$

Z-transform properties

Sequences $x[n], y[n]$	Transforms $X(z), Y(z)$	ROC	Property
$ax[n] + by[n]$	$aX(z) + bY(z)$	ROC contains $R_x \cap R_y$	Linearity
$x[n - n_d]$	$z^{-n_d} X(z)$	ROC = R_x	Time shift
$z_0^n x[n]$	$X(z/z_0)$	ROC = $ z_0 R_x$	Frequency scale
$x^*[-n]$	$X^*(1/z^*)$	ROC = $\frac{1}{R_x}$	Time reversal
$nx[n]$	$-z \frac{dX(z)}{dz}$	ROC = R_x	Frequency diff.
$x[n] * y[n]$	$X(z)Y(z)$	ROC contains $R_x \cap R_y$	Convolution
$x^*[n]$	$X^*(z^*)$	ROC = R_x	Conjugation

Common z-transform pairs

Sequence	Transform	ROC
$\delta[n]$	1	All z
$u[n]$	$\frac{1}{1-z^{-1}}$	$ z > 1$
$-u[-n-1]$	$\frac{1}{1-z^{-1}}$	$ z < 1$
$\delta[n-m]$	z^{-m}	All z except 0 or ∞
$a^n u[n]$	$\frac{1}{1-az^{-1}}$	$ z > a $
$-a^n u[-n-1]$	$\frac{1}{1-az^{-1}}$	$ z < a $
$na^n u[n]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$ z > a $
$-na^n u[-n-1]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$ z < a $
$\begin{cases} a^n & 0 \leq n \leq N-1, \\ 0 & \text{otherwise} \end{cases}$	$\frac{1-a^N z^{-N}}{1-az^{-1}}$	$ z > 0$
$\cos(\omega_0 n) u[n]$	$\frac{1-\cos(\omega_0)z^{-1}}{1-2\cos(\omega_0)z^{-1}+z^{-2}}$	$ z > 1$
$r^n \cos(\omega_0 n) u[n]$	$\frac{1-r\cos(\omega_0)z^{-1}}{1-2r\cos(\omega_0)z^{-1}+r^2 z^{-2}}$	$ z > r $

Other

$$\sum_{n=0}^{N-1} a^n = \frac{1-a^N}{1-a} \quad \text{for } a \neq 1.$$